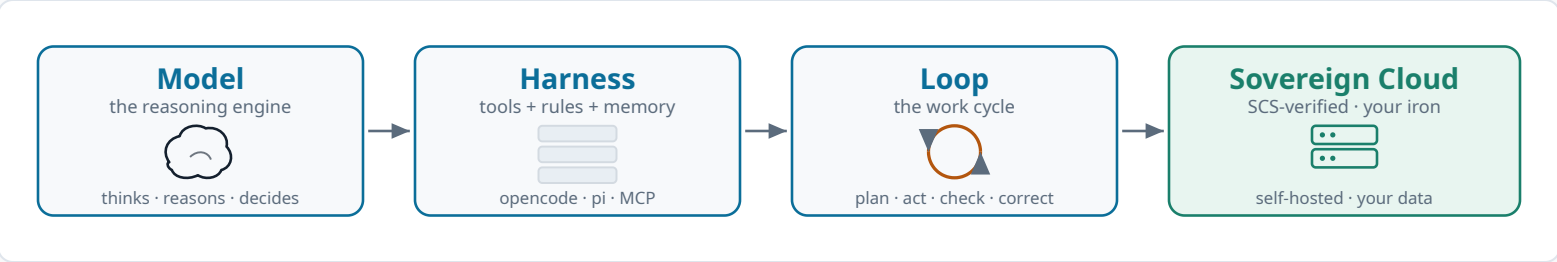
SOVEREIGN CLOUD DAYS 2026 · WORKSHOP STAGE

Agentic AI meets Sovereign Cloud

openEduSuite on SCS-verified legacy bare-metal hardware

Sun, 25.10.2026 · 11:15 · 60 min incl. Q&A

Tobias Weiß



CONTEXT · PROJECT COMPARISON

Yaook vs. OSISM — two paths to sovereign OpenStack

Yaook · yaook.cloud
OpenStack **IN** Kubernetes (CRDs)
~12 core services · kubectl + yaookctl
K8s-native, declarative, rolling releases
Metal Controller + Netbox · CUE validation

OSISM · osism.tech
OpenStack **ALONGSIDE** Kubernetes (Ansible)
~25 services · Web UI + Manager CLI (38 cmds)
Full ops suite · 15+ security/monitoring tools
Metalbox + Ironic + SONIC · CUE (adopted from Yaook)

SCS: OSISM is the reference implementation; Yaook is an SCS member. Both Apache 2.0 · European.

MOTIVATION · 3 MIN

Legacy hardware isn't junk

- Data centers have **decommissioned servers** — often fully functional.
- A sovereign, AI-capable setup runs on **written-off iron**.
- 3 × Huawei RH2288E V2 (~2013) → SCS-verified K3s cluster.

> openEduSuite: HFD Digital Sovereignty Ideas Competition — winner 2026-10-06.



- Whole sovereign workplace — identity, mail, collaboration, delivery
- Real FOSS components
- Verified against SCS (6/6 KaaS standards PASS)

WS **Who runs “legacy hardware” in production themselves?**

MOTIVATION · THE PIVOT · 3 MIN

From OpenStack to bare-metal K3s

OpenStack / OVN on VMs

OVN 25.03.1 gateway bug
FIPs dead, DHCP broke

"Reboot destroys the fix"

Pivot

Bare-metal K3s (SCS-verified)

3 nodes · Flannel · MetalLB · Ceph
6/6 KaaS standards PASS · 444/444 CNCF

fewer layers, fewer failure points

What broke in OVN 25.03.1: gateway redirect bug · floating IPs (FIPs) unreachable · DHCP leases lost · OVN northd/southbound DB + OVS bridges on every node · "reboot destroys the fix" — every node restart re-triggered it.

The VM/SDN abstraction was too heavy for 3 nodes — bare metal is leaner.

HANDS-ON · INSTALL · 2 MIN

Install your tool — pick one

Beginners · opencode

```
curl -fsSL https://opencode.ai/install | bash
# Or: npm install -g opencode-ai
# Or: bun add -g opencode-ai (much faster)
# Verify: opencode --version
# Start: cd <project> && opencode
```

Advanced · pi.dev

```
curl -fsSL https://pi.dev/install.sh | sh
# Or: npm i -g @earendil-works/pi-coding-agent (Node 22.19+)
# Or: bun add -g @earendil-works/pi-coding-agent (much faster)
# Verify: pi --version
# Start: cd <project> && pi
```

Both are free to start — no paid API key required. Using **bun** gives a noticeable speedup.

HANDS-ON · CHOOSING A MODEL · 3 MIN

Start for free — Big Pickle or Beam

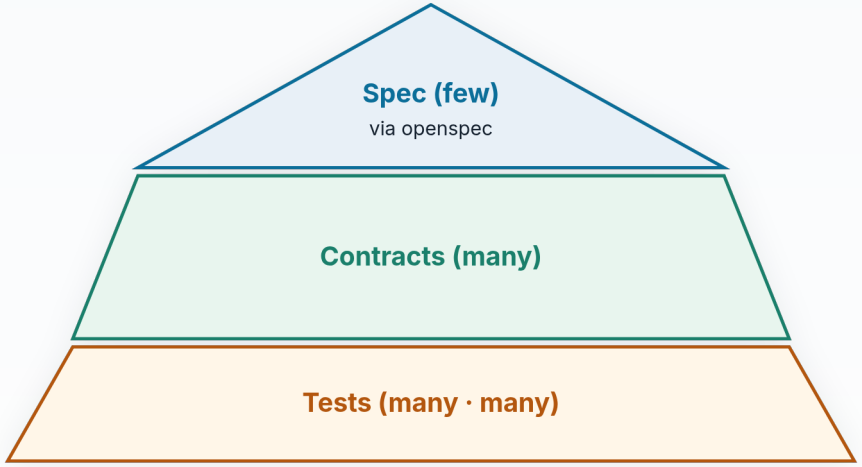
- **opencode (beginners):** `/connect` → **OpenCode Zen** → pick **Big Pickle** or **Beam Free**
- **pi.dev (advanced):** `/login` → provider **opencode**, then `/model` → `big-pickle` or `beam-free`
- Alternatives: any provider via `/connect` (75+) or your own API key

Both are free on OpenCode Zen — enough for agentic experiments, no cost risk.

⚠️ **Privacy — read before you paste.** Free cloud models differ in how they handle your data. **Big Pickle** is a stealth model; its provider **may use your input to train**. **Beam Free** (Reflection's open-weight 501B model, 1M-token context) does **not** train on your data (retains it 30 days for safety). **Never paste private, customer, or unpublished data into a model that trains on user input.** For sensitive work, run a local model (Ollama, `ollama run beam`) or your own sovereign endpoint.

QUALITY · SPEC-DRIVEN · 3 MIN

Spec → Contract → Test pyramid



SPEC
(few) — the single source of truth, written with openspec.

CONTRACTS
(many) — derived from the spec, agreed between producer and consumer.

TESTS
(many · many) — fast, cheap, run on every change; the broad base.

CLUSTER · HARDWARE · 3 MIN

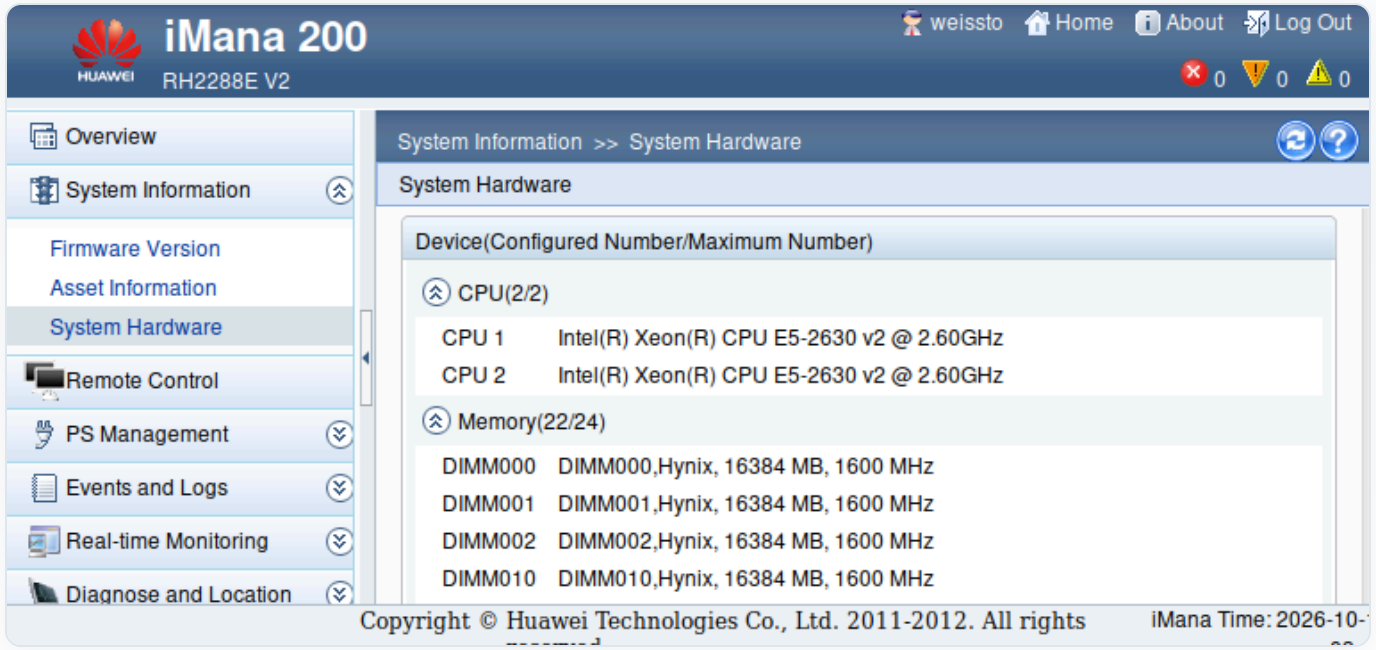
3 × Huawei RH2288E V2 — no GPU

CPU
2× Xeon E5-2600 v2 (Ivy Bridge-EP) · 6–12 cores

RAM
24× DDR3 ECC RDIMM · max 768 GB

STORAGE
6× 512 GB SSD · Ceph 8 OSDs · Repl. 3

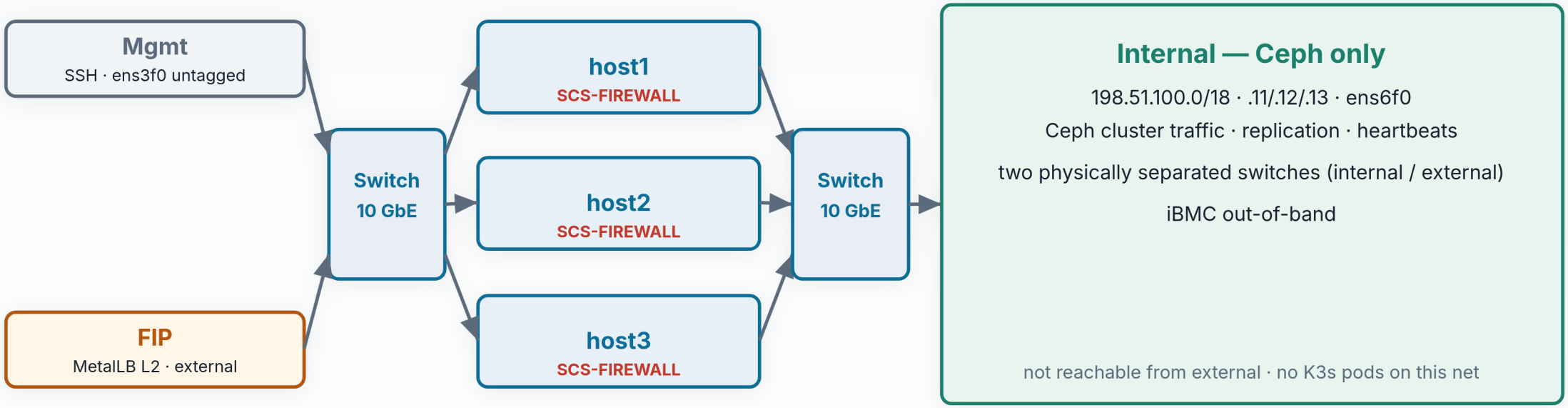
NETWORK
2× Intel X540-AT2 10 GbE



iBMC iMana — hardware view (host1/host2/host3)

CLUSTER · NETWORK · 3 MIN

Three networks, one firewall



Left: Mgmt + FIP (external) → Switch 10G → 3 nodes (SCS-FIREWALL) → Switch 10G → Right: Internal (Ceph only) · 2 phys. switches · iBMC OOB

CLUSTER · K3S STACK · 4 MIN

The stack — bare-metal in production

openEduSuite Workloads (edu · staff · students · sme · llm)

ArgoCD (GitOps) · K8up/Restic (backup → SeaweedFS S3)

Ceph 18.2.8 Reef — 8 OSDs, RBD + CephFS, Repl. 3

Flannel VXLAN · MetalLB L2 · HAProxy Ingress

K3s v1.36.3+k3s1 · Single-server · embedded datastore

Ubuntu 24.04 LTS · containerd 2.3.3 · runc 1.4.3

3 × Huawei RH2288E V2 (bare metal)

CLUSTER · SCS-VERIFIED · 3 MIN

6/6 KaaS standards PASS — measurable

STANDARD	TEST	RESULT
SCS-0201 CNCF conformance	`cncf-k8s-conformance` (444/444 specs)	✅ PASS
SCS-0210 version policy	`version-policy-check` (v1.36.3 in window)	✅ PASS
SCS-0211 default storage class	manual (ceph-rbd, repl. 3)	✅ PASS
SCS-0214 node distribution	manual (3 nodes, zones a/b/c)	✅ PASS
SCS-0217 pod security	manual (8 NS baseline; 2 system NS privileged)	✅ PASS
SCS-0219 KaaS networking	`kaas-networking-check` (37 NetworkPolicies)	✅ PASS

Open: OIDC as 2nd auth · Pod-Security on more NS · Loki/Trivy CI · Ceph at rest · GDPR clearance.

Criteria & tests: github.com/SovereignCloudStack/standards

CONTRIBUTIONS · SCS · 2 MIN

PRs I sent to SCS repositories

REPOSITORY	PR	TITLE	STATUS
SovereignCloudStack/standards	#1267	feat: add bare-metal K3s implementation notes for SCS-0217	closed
SovereignCloudStack/docs	#397	feat: add bare-metal K3s deployment guide	closed

Both PRs document how the bare-metal K3s deployment satisfies SCS-0217 (KaaS networking) and the wider KaaS standard set.

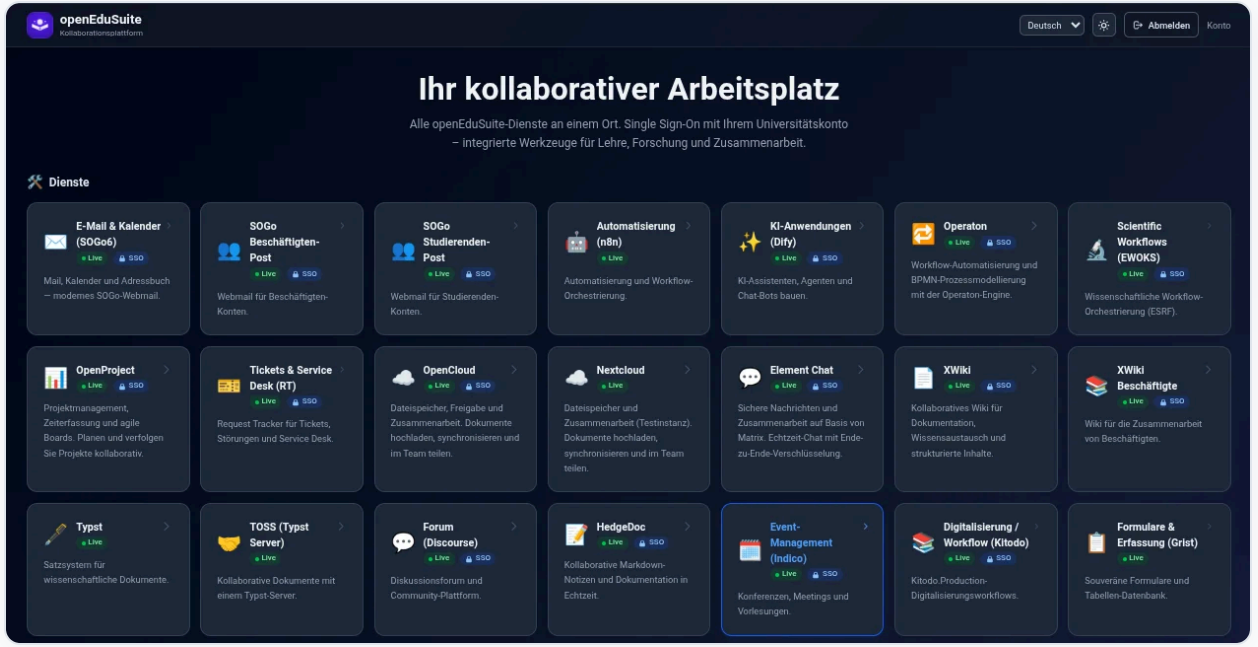
OPENEDUSUITE · WORKLOAD · 4 MIN

The dimensions of openEduSuite

DIMENSION	COMPONENTS
Tenants — who	edu · staff · students · sme · llm · predictive-agent
Identity & access	Keycloak / OIDC · OpenLDAP
Communication	Postfix · Stalwart · DKIMpy · Matrix
Collaboration	Nextcloud · XWiki · SOGo · OpenProject · Typst · Toss
Workflows	Dify · n8n · Ewoks
Delivery	Own Nix builds on GHCR → local Zot registry · ArgoCD

OPENEDUSUITE · LIVE DEMO · 4 MIN

Cluster, login, agent trace



openEduSuite — bird's-eye view of the platform

- Dashboard: 3 nodes · Ceph HEALTH OK (8/8) · SCS board
- Login via Keycloak/OIDC → agentic workflow

> **Fallback:** Screenshots (Grafana, KVM, open-webui) + stack diagram if the demo stalls.

SELF-CHECK · EU CLOUD SOVEREIGNTY SCORE · 2 MIN

73.5 % Sovereignty Score — SEAL-1

Method: **EU Cloud Sovereignty Framework** (official calculator, Jun 2026) — weighted **Sovereignty Score** + **SEAL** level (weakest link across all 48 criteria).

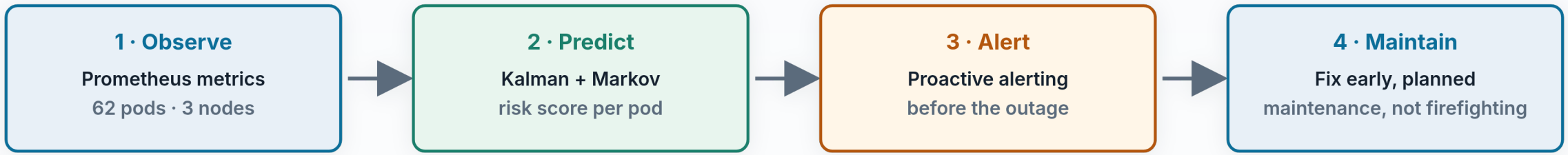
OBJECTIVE	W	SCORE	SEAL
SOV-1 Strategic	20 %	**94 %**	SEAL-4
SOV-4 Operational	15 %	**83 %**	SEAL-3
SOV-6 Technology	15 %	**80 %**	SEAL-3
SOV-3 Data & AI	10 %	75 %	SEAL-1
SOV-2 Legal & Jurisdictional	10 %	71 %	SEAL-1
SOV-7 Security & Compliance	15 %	64 %	SEAL-1
SOV-5 Supply Chain	10 %	50 %	SEAL-1
SOV-8 Environmental	5 %	19 %	SEAL-1

SOV = Sovereignty objective · W = weight · SEAL = Sovereignty Assurance Level (weakest link) · EAL = Evaluation Assurance Level · PUE = Power Usage Effectiveness · CSRD = Corporate Sustainability Reporting Directive

High score — low SEAL (weakest link): non-EU hardware (Huawei/Intel) · no EAL cert (SOV-7) · no PUE/CSRD reporting (SOV-8). Levers: EU hardware provenance · EAL cert · PUE reporting.

OPENEDUSUITE · AGENTIC WORKFLOW · 4 MIN

Cluster maintenance in 4 steps

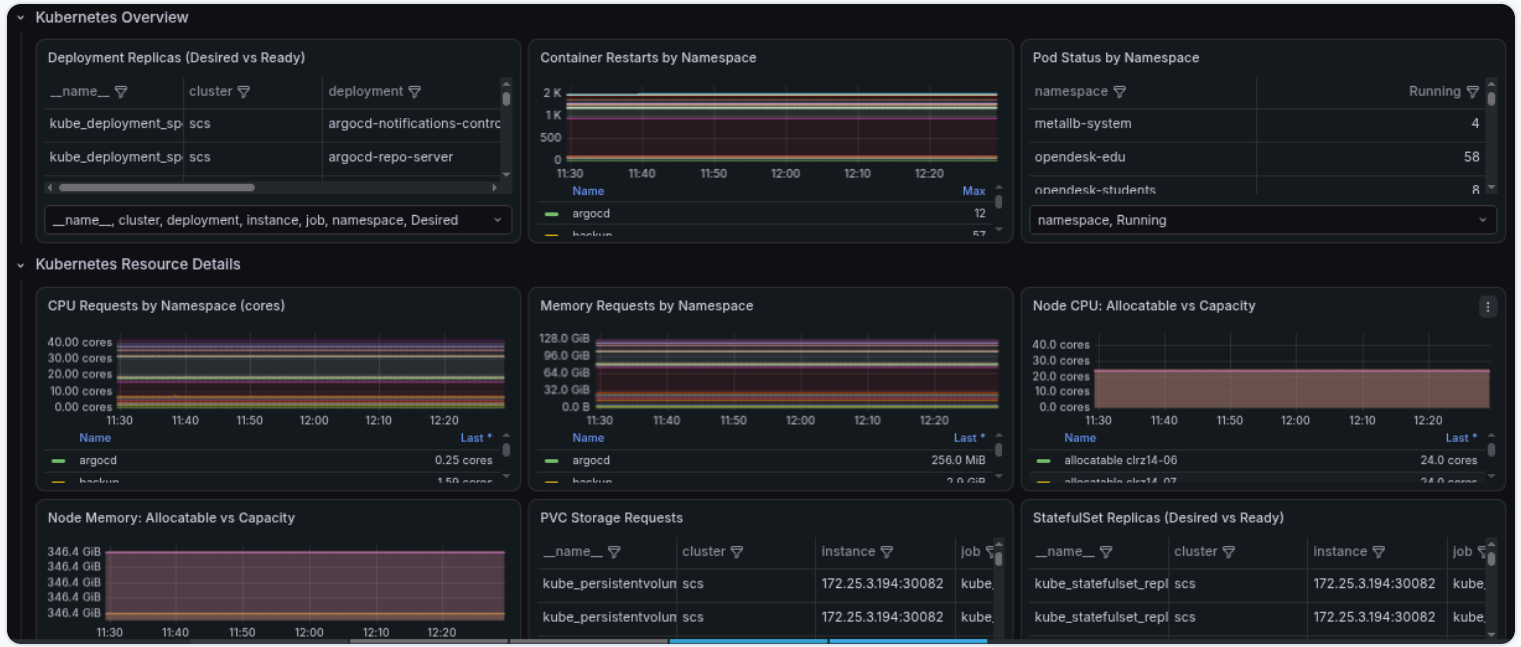


predictive-agent v8.11-nix (own tenant): Risk scoring (Kalman + Markov) for 62 pods — alerts before users notice.

Kalman filter: smooths noisy Prometheus metrics into a state estimate per pod. **Markov chain:** predicts the next health state (OK → degraded → down) from the smoothed state. Combined → risk score before the user notices.

CLUSTER · OPERATIONS · 2 MIN

What we learned in operation



Grafana K3s dashboard (7 dashboards, predictive-agent)

- Stale digests, default-deny NetworkPolicies, OIDC migration — all resolved and documented
- Updates: Ceph-first, rolling, rollback script · backup: K8up → SeaweedFS S3

Stale digests: pinned image digests drifted from the registry; fixed with digest refresh + ArgoCD sync. **Default-deny NetPolicies:** 37 NetworkPolicies initially blocked DNS/DB traffic; fixed with explicit allow rules. **OIDC migration:** Keycloak/OIDC rolled out across all services. **Updates:** Ceph-first rolling upgrades with a rollback script. **Backup:** K8up → SeaweedFS S3.

LESSONS · 3 MIN

Five takeaways

- **Start small, learn fast** — OpenStack didn't fit the scale.
- **Bare metal instead of a VM layer** — less abstraction, more control.
- **Constraints are features** — quantization beats buying GPUs.
- **“Verified” convinces** — SCS tests open doors, with honest gap reporting.
- **Document like a madman** — runbooks, DPIAs, checklists.

Q&A · 10 MIN

Questions & discussion

- Architecture, rebuilding it, SCS/openEduSuite licensing topics
- Workshop element: opencode install + the prompts — try them at home
- Contact: tobias-weiss.org · graphwiz.ai · ki-kompetenz-training.org

> Thanks — questions welcome, and let’s keep the conversation going.